

FreshChain Deployment and Setup Guide

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Audience: SAP BTP administrators, platform engineers, CAP developers, and solution owners

1. Purpose

This guide explains exactly how to deploy the FreshChain production demo suite to an SAP BTP Cloud Foundry subaccount with SAP AI Core and SAP AI Launchpad. It covers the CAP backend, HANA persistence, managed HTML5 application deployment, Destination/XSUAA setup, AI Core runtime setup, Launchpad connection, validation, and troubleshooting.

FreshChain uses SAP AI Core for training, deployment, and inference. SAP AI Launchpad is the operator console for AI Core scenarios, executions, deployments, logs, and metrics. The CAP application does not perform local inference and does not use a local model fallback. If SAP AI Core inference is unavailable, FreshChain records failed telemetry and returns an operational error.

2. Solution Components

Component	Purpose	Deployment source
<code>freshchain-srv</code>	CAP Node.js service exposing OData and ingestion endpoints	<code>gen/srv</code> from <code>cds build</code>
<code>freshchain-db-deployer</code>	HANA HDI deployer for CDS persistence artifacts	<code>gen/db</code>
HTML5 apps	Overview, Intelligence, Operations, Monitoring, Master Data, Admin	<code>app/freshchain-*</code>
<code>freshchain-html5-repo-host</code>	Stores static HTML5 app content	MTA resource
<code>freshchain-destination</code>	Runtime destination and HTML5 runtime configuration	MTA resource
<code>freshchain-auth</code>	XSUAA application security	<code>xs-security.json</code>
<code>freshchain-ai-core</code>	SAP AI Core service binding consumed by CAP	MTA resource
SAP AI Launchpad	AI operations console connected to AI Core	Subaccount subscription
AI Core scenario assets	Scenario and executable definitions	<code>ai-core/*.yaml</code>
AI model image	Training and serving container	<code>ml/Dockerfile</code>
Dataset upload persistence	Stores uploaded CSV ZIP packages, checksums, validation summaries, and import lineage	CAP/HANA entity <code>DatasetUploads</code>

3. Prerequisites

3.1 Local workstation

Install and verify:



Required tools:

- Node.js compatible with the CAP dependencies in `package.json`.
- Cloud Foundry CLI with access to the target org and space.
- MultiApps build tool (`mbt`) for MTAR generation.
- Docker or another OCI-compatible image builder.
- Access to a container registry that SAP AI Core can pull from.

3.2 SAP BTP subaccount

The target subaccount must have:

- Cloud Foundry environment enabled.
- A Cloud Foundry org and space.
- Entitlements for:
 - SAP HANA Cloud / HDI container service plan `hdi-shared`
 - Authorization and Trust Management service `xsuaa`
 - Destination service `lite`
 - HTML5 Application Repository `app-host`
 - SAP AI Core service `aicore`, plan `standard`
 - SAP AI Launchpad subscription
 - A role collection that allows administration of AI Launchpad connections. SAP documents the Launchpad AI Core connection flow as requiring the `connections_editor` role or equivalent role collection.

SAP references:

- SAP AI Core service instance setup: <https://help.sap.com/docs/sap-ai-core/sap-ai-core-service-guide/create-service-instance>
- SAP AI Core service key usage: <https://help.sap.com/docs/sap-ai-core/sap-ai-core-service-guide/use-service-key>
- SAP AI Launchpad connection to AI Core: <https://help.sap.com/docs/ai-launchpad/sap-ai-launchpad/add-connection-to-sap-ai-core>
- HTML5 Application Repository overview: <https://help.sap.com/docs/build-work-zone-advanced-edition/sap-build-work-zone-advanced-edition/what-is-html5-application-repository>

4. Repository Verification

From the repository root:



Expected result:

- `npm run check` prints `UI artifact validation passed`.
- `npm test` passes all tests.
- Tests mock SAP AI Core and verify that inference fails closed when AI Core is unavailable.

5. Build and Push the AI Image

FreshChain includes a model server and training entry point under `ml/`.

Build the image:



Push it:



The same image supports:

- Training command: `python /app/train.py`
- Serving command: `uvicorn serve:app --host 0.0.0.0 --port 9000`

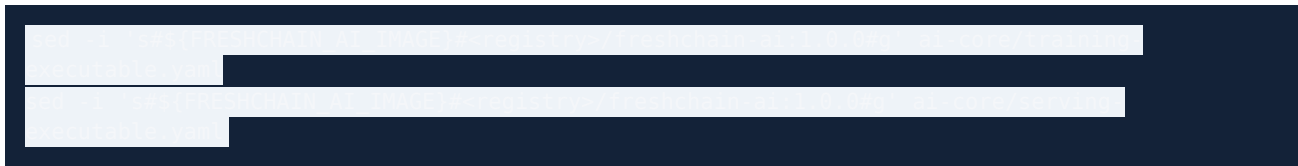
Confirm the image path is reachable by SAP AI Core. If your registry requires credentials, configure the registry secret in AI Core according to your organization's AI Core operations standard.

6. Prepare AI Core Scenario Assets

FreshChain ships these AI Core assets:

File	Purpose
<code>ai-core/scenario.yaml</code>	Scenario <code>freshchain-intelligence</code>
<code>ai-core/training-executable.yaml</code>	Training executable <code>freshchain-train</code>
<code>ai-core/serving-executable.yaml</code>	Serving executable <code>freshchain-serve</code>

Replace the placeholder image reference in both executable files:



Use resource group:



Use IDs:



Register the scenario and executables using your organization's preferred SAP AI Core workflow, such as AI Core API, AI Core toolkit, or GitOps onboarding process. After registration, verify the scenario and executable names are visible in SAP AI Launchpad after the Launchpad connection is created.

7. Configure BTP and Cloud Foundry

Log in:



Confirm service marketplace entries:



The MTA creates these managed services:



Important: SAP AI Launchpad is a subaccount subscription. It is not deployed as an MTA module. Subscribe to Launchpad in the BTP cockpit before the business team starts validating AI operations.

8. Build the MTAR

Run:



Expected output:



The build performs:

- `npm ci`
- `npx cds build --production`
- CAP service packaging
- HANA deployer packaging
- HTML5 app zip generation
- HTML5 content packaging

Existing CAP warnings about cross-service navigation properties in `IntelligenceService` can be reviewed, but they do not block deployment.

9. Deploy to Cloud Foundry

Deploy:



Validate:



Expected:

- `freshchain-srv` is started.
- `freshchain-db-deployer` has completed successfully.
- `freshchain-ai-core` exists and is bound to `freshchain-srv`.
- HTML5 app content is uploaded to the HTML5 Application Repository.

10. Enable Dataset Uploads

Dataset upload is part of the CAP service and requires no separate object store for the production demo. Uploaded ZIP packages are persisted in HANA through the `DatasetUploads` entity and linked to the imported `MLDatasets` record after import.

Operational limits:

- Maximum decoded ZIP size: 25 MB.
- HTTP JSON body limit in the CAP server: 40 MB.
- Required files: `sensor_readings.csv` and `sales_observations.csv`.
- Optional master data files can create stores, zones, products, batches, and inventory placements before importing observations.

Validation and import flow:

- Data science users upload the ZIP from FreshChain Intelligence.
- Users can download a generated starter ZIP from the Template button if they need the expected CSV layout.
- CAP stores the package content, SHA-256 checksum, filename, MIME type, and upload timestamp.
- CAP validates required files, columns, date formats, numeric fields, duplicate sensor message IDs, and master-data references.
- The UI shows row counts and validation errors.
- Users explicitly choose Import after successful validation.
- CAP imports master data, sensor readings, sales observations, and one `MLDatasets` lineage record.
- Imported packages are retained and cannot be deleted, preserving training lineage.

Post-deploy validation:



The response should return an OData collection. Use the FreshChain Intelligence app for normal upload operations because it calls the CAP OData actions and avoids direct browser HTTP code in the UI.

If the AI Core service creation fails, confirm the subaccount entitlement and plan name. The MTA uses service `aicore` and plan `standard`.

10. Create AI Core Service Key for Launchpad

Create a service key:



Keep the output secure. It contains credentials used by SAP AI Launchpad and API clients.

You need these values:

- `AI_API_URL`
- `OAuth url`
- `clientid`
- `clientsecret`

SAP documents that the service key provides URLs and credentials for AI Core access, and that `clientid`, `clientsecret`, and `url` are used to obtain an OAuth token.

11. Connect SAP AI Launchpad to AI Core

In SAP AI Launchpad:

- Open the Workspaces app.
- Choose Add.
- Create an AI API connection.
- Upload the AI Core service key JSON or enter fields manually.
- If using client secret credentials, enter:

- `AI_API_URL`

- `url`

- `clientid`

- `clientsecret`

- Save the connection.

Expected:

- The new connection appears in Workspaces.
- Scenario `freshchain-intelligence` is visible.
- Executables `freshchain-train` and `freshchain-serve` are visible after scenario asset registration.

12. Runtime Configuration

The CAP service reads SAP AI Core credentials from `VCAP_SERVICES` through the bound `freshchain-ai-core` instance.

Optional environment overrides:



Defaults already match the shipped assets:

Variable	Default
<code>AICORE_RESOURCE_GROUP</code>	<code>freshchain-demo</code>
<code>AICORE_SCENARIO_ID</code>	<code>freshchain-intelligence</code>
<code>AICORE_TRAINING_EXECUTABLE_ID</code>	<code>freshchain-train</code>
<code>AICORE_SERVING_EXECUTABLE_ID</code>	<code>freshchain-serve</code>

13. HTML5 Apps and Work Zone

FreshChain deploys HTML5 app content and destination content. The design assumes managed HTML5 runtime and Work Zone/Launchpad-style access, not a custom standalone approuter.

The apps are:

App	Purpose
FreshChain Overview	Executive and operational summary
FreshChain Intelligence	AI Core training, deployment, scoring, quality, telemetry
FreshChain Operations	Alert workflow
FreshChain Monitoring	Zone health and active alert monitoring
FreshChain Master Data	Stores, zones, products, batches, sensors
FreshChain Administration	Thresholds and ingestion errors

In Work Zone, add the HTML5 applications to the desired site/catalog/group according to your tenant's content management process. Use the app titles and semantic objects defined in each app manifest.

14. Validation Checklist

14.1 CAP and OData

Find the service route:



Open:

Expected:

- Metadata loads.
- Authentication behavior matches the XSUAA setup.
- `IntelligenceService` exposes `startTraining`, `activateDeployment`, `refreshTrainingRun`, `refreshDeployment`, and `scoreLatest`.

14.2 AI Core lifecycle

From FreshChain Intelligence:

- Confirm at least one dataset exists.
- Choose Start AI Core Training.
- Open SAP AI Launchpad and verify a training execution appears.
- Refresh training status in FreshChain.
- Activate latest deployment.
- Verify the deployment in SAP AI Launchpad.
- Refresh deployment status until online.
- Score a zone.
- Confirm inference telemetry shows AI Core reached.

14.3 Fail-closed behavior

If AI Core inference is unavailable:

- `scoreLatest` returns an error.
- `InferenceTelemetry` records status `FAILED`.
- `aiCoreUnavailable` is true.
- No synthetic prediction is generated.
- No local model is used.

15. Troubleshooting

Symptom	Likely cause	Resolution
<code>aicore</code> service cannot be created	Missing entitlement or wrong plan	Add SAP AI Core entitlement and confirm plan <code>standard</code>
Launchpad cannot connect to AI Core	Wrong service key values or missing role	Use AI Core service key and assign Launchpad connection role
Scenario missing in Launchpad	AI Core assets not registered	Register <code>ai-core/*.yaml</code> in resource group <code>freshchain-demo</code>
Training action fails with missing binding	<code>freshchain-srv</code> not bound to AI Core	Confirm <code>freshchain-ai-core</code> binding and restage app
Deployment has no endpoint	AI Core deployment not online	Check Launchpad deployment logs and image pull status
Image pull fails	Registry credentials missing	Configure registry access for SAP AI Core
HTML5 apps not visible	Work Zone content not assigned	Add apps to Work Zone site/catalog/group

Symptom	Likely cause	Resolution
OData route fails	Destination or auth misconfiguration	Validate Destination service and XSUAA bindings

16. Production Handover Checklist

- MTAR deployed successfully.
- AI image built and pushed.
- AI Core scenario and executables registered.
- Launchpad connected to AI Core.
- FreshChain apps available in Work Zone.
- Role collections assigned to administrators, data scientists, store managers, and integration users.
- Training execution visible in Launchpad.
- Deployment online in Launchpad.
- FreshChain scoring creates predictions and recommendations from AI Core output.
- Failed inference telemetry is visible and no local fallback is active.